

Development of Parallel Stance Check System for Skiing スキーマの平行スタンスチェックシステムの開発

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1. Background and Goal

Parallel turn (PT) is performed with both skis parallel, and parallel stance (PS) is the posture during such turns with approximately parallel lower limbs [1]. PS is important element in PT in skiing. However, checking PS has traditionally relied on instructors' visual inspection. Therefore, in this study, I developed a system that automatically checks PS using recordable skiing videos. PS quality will be more easily visualized, allowing skiers and instructors to check PS more easily.

2. Approach/Methodology

This system allows users to upload videos and outputs PS detection results after going through the process in Fig. 1. .

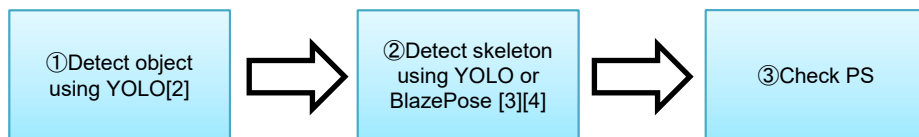


Fig. 1 Main system operations.

The problems encountered and their solutions are shown below.

① How to track target

For the required object detection model, the accurate and fast 'yolo11n.pt' model was adopted [5]. In general, object detection model may fail to recognize the target when other skiers are visible in the video or when the target appears small due to a long shooting distance. Therefore, a Region of Interest (ROI) was set as shown in Fig. 2. for object detection and tracking. When the skier is filmed from below, the skier appears larger in the camera view toward the end of the video. By rewinding the video from that point, 'yolo11n.pt' can obtain the initial ROI value.

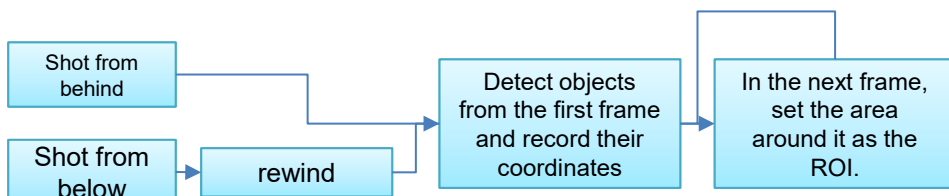


Fig. 2 Procedure to set ROI.

② What model should be used for skeleton detection?

'yolo11n-pose.pt', which can obtain 2D skeletons and highly accurate skeletal information [5], and 'pose_landmarker_full.task' of BlazePose, which can obtain 3D skeletons [6], are considered. The accuracy and stability of both models are tested to determine which model to use.

③ How will algorithm check PS?

PS checking algorithm will extract key points of both shins and calculate the angle between them (Fig. 3.).

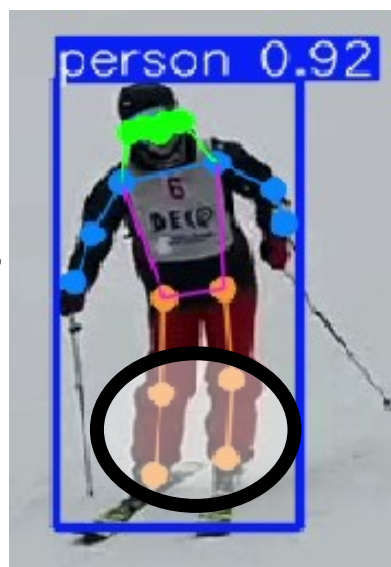


Fig. 3 Key points to check PS.

3. Current Results

To confirm whether the proposed method can track the target, Object detection test performed by using 'yolo11n.pt' on a skiing video taken from below. For comparison, result of a test performed under normal conditions is also shown. Fig. 4. shows the results. In the latter case, the target is small, so detection fails in the first half of the video, but in the former case, detection is almost successful.

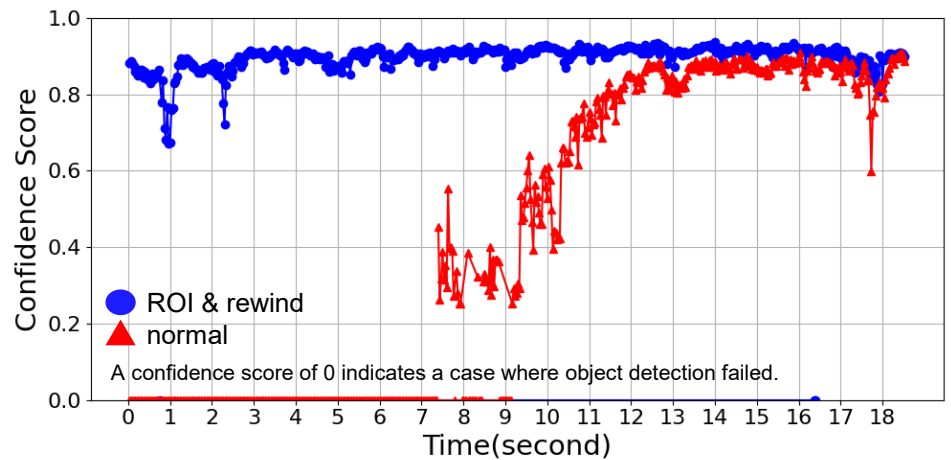


Fig. 4 Results of detection test of the ROI & rewind method. Results of the normal playback case are also plotted.

To confirm that PS checking algorithm works correctly, it was applied to skeleton detected by 'pose_landmarker_full.task' [7]. Fig. 5. shows the result. The angle fluctuates significantly within one second, which is unrealistic for human motion considering that FPS is 30. This indicates that the current algorithm fails to calculate the angle correctly. The algorithm will be improved by applying smoothing to remove noise and by focusing on other parts of the foot. 'yolo11n-pose.pt' will be tested.

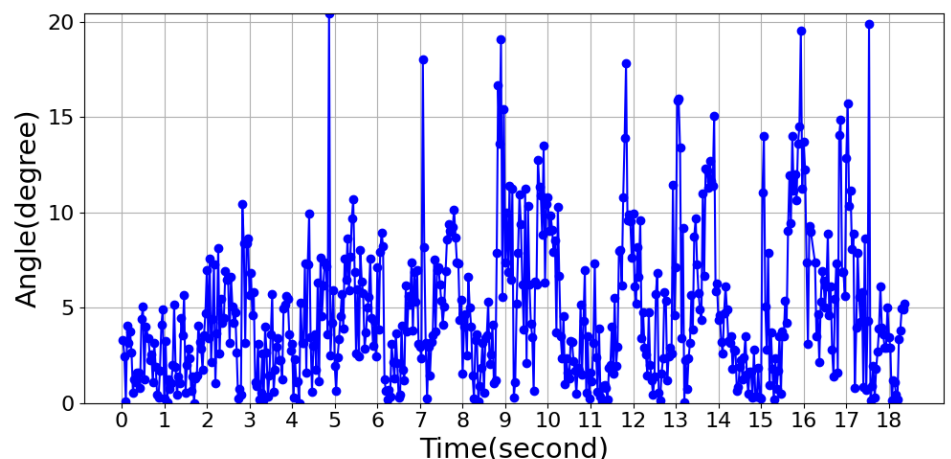


Fig. 5 Shins angles computed from 'pose_landmarker_full.task' results. At 16.4 second, 'yolo11n.pt' is failed to detect the target object.

4. Remaining Tasks and Tentative Schedule

Schedule	10	11	12	1	2
Complete to parallel stance checking algorithm					
Decide which skeleton detection model to use					
Develop UI					
Write graduate thesis & prepare for presentation					

5. References

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